Phase 2 final analysis: clinically realistic cases

2026-10-19 · Prepared by engineering · Status: built and oracle-verified; **clinical content awaiting oncologist sign-off**

No v2 score should be reported as clinical performance until an oncologist signs off the reference answers.

1. Summary

OncoWorld v1 had a sound harness: Docker sandboxes, a hidden verifier, and narrative and FHIR versions of each case. But its cases were short, pre-interpreted bullet lists, each built around a single teaching point, and its regex graders scored wording more than reasoning. Phase 2 rebuilds the same five DLBCL patients as realistic charts and adds a five-layer longitudinal journey. Grading now uses atomic rubric items with severities, so a single dangerous error fails the case.

- **Realism:** 10-14 dated documents per case (referral, consult, radiology, pathology with FISH addendum, labs with reference ranges, medications, distractor comorbidities). Narrative and FHIR versions are generated from one source and cannot disagree.
- **Decision-changing twists** taken from real practice: occult hepatitis B, anthracycline cardiotoxicity, tumour-lysis risk with rising creatinine and NSAID use, a double-hit FISH addendum that arrives after a plan already exists, and a contraindicated contraceptive.
- **The v1 reference answers fail the v2 bar.** Graded against the v2 rubrics, three score 0.00 because they miss a dangerous item (HBV prophylaxis, cardiac function, TLS prophylaxis), and the other two score 0.47 and 0.43. That gap is what the new cases test.
- **Verified:** 15 self-test suites pass (37/37 case-level verdicts as expected). Harbor oracle runs: v1 11/11 and v2 11/11 trials at full score (v2 includes the 5-step journey).
- **Not yet done:** oncologist sign-off, a funded-model baseline, and measuring how well the LLM judge agrees with the oncologist.

2. Assessment of v1 (the starting point)

Workflow in use

- Write a short bullet-point case (history, labs, imaging, pathology).
- Generate a matching FHIR R4/mCODE bundle from a hard-coded Python dictionary.
- Write a gold answer, a rubric and a regex grader per case (11 copies, 6 different versions).
- Run oracle, then Gemini CLI / OpenCode models, mostly on free tiers.
- Read review.json, patch regexes when correct wording was marked wrong, and add the answer to the self-tests.

What blocked trust

Problem	Effect
No valid baseline under the 2026-10-13 weights; many runs failed on 429/503 errors and setup timeouts	No model score could be reported
No physician sign-off; the spleen staging rule was still marked 'to confirm'	Gold answers were engineering opinion
Regex graders patched 6 times and still brittle to paraphrase and negation	Scores partly measured wording
11 copy-pasted graders	Fixes did not propagate between cases

Where v1 departed from real practice

- No past history, medications, allergies or reference ranges; findings arrived pre-interpreted ('LDH elevated').

- The narrative and FHIR versions disagreed: the narrative said 'elevated' while the FHIR bundle omitted the range on purpose.
- 'Missing information' was manufactured by withholding PET findings a real report would state.
- Leading questions named the concept being tested ('double-expressor or double-hit?').
- No pre-treatment work-up, supportive care or patient goals, though these drive real treatment decisions.
- Gold gaps: no FLYER for limited stage (0005); no WHO-HAEM5 name accepted (0004); an IPI wording slip (0001); no prephase, TLS or methotrexate-feasibility content (0003).
- A single snapshot in time, while real oncology unfolds over months.

3. What Phase 2 changed

Area	v1	v2
Chart	4 short text files	10-14 dated documents with reference ranges, computed H/L flags, past history, meds, allergies, distractors
Formats	Hand-built separately; disagreed	One source (cases_src/<case>/case.toml) generates both narrative and FHIR
Prompt	Leading per-concept questions	'Consulting hematologist: write the tumor-board assessment and plan'
Answer keys	8 keys	11 keys, adding pretreatment_workup, supportive_care, fitness_and_goals
Grading	Regex per case, points minus penalties	Atomic must / should / must-not items with minor / important / dangerous severity; a dangerous error means reward 0
Judge	None	Optional LLM judge that must quote the answer verbatim (invented quotes are rejected); regex fallback
Safety checks	None	Build fails on answer leakage into the chart, pathology inconsistency, or a gold answer below 1.0
Time	One snapshot	5-layer journey as a native Harbor multi-step task

4. Results

v1 reference answers graded against the v2 rubrics

Same patient, same core facts, but the v2 chart adds the real-world twist. A clinically complete plan must now handle it.

Case	v2 twist	v1 gold reward	Dangerous errors
dlbcl_0001	Occult hepatitis B (HBsAg-, anti-HBc+) before rituximab	0.00	hbv_prophylaxis
dlbcl_0002	Prior doxorubicin 240 mg/m² + chest RT; LVEF 62% → 45%	0.00	tx_cardiac_aware_no_full_dox, mn_standard_dox_ignores_heart
dlbcl_0003	TLS risk: uric acid and creatinine rising, on ibuprofen; no CNS staging	0.00	sup_tls_prophylaxis
dlbcl_0004	Late FISH addendum: double-hit, after an R-CHOP plan was charted	0.47	none (fails on points)
dlbcl_0005	Migraine with aura on estrogen pill; no pregnancy test; avoid neck RT	0.43	none (fails on points)

Harbor oracle runs (Docker, 2026-10-19)

v1 task (current weights)	Reward
breast_0001_fhir	1.00
dlbcl_0001_fhir	1.00
dlbcl_0001_narrative	1.00
dlbcl_0002_splenic_stage_iii_fhir	1.00
dlbcl_0002_splenic_stage_iii_narrative	1.00
dlbcl_0003_renal_adrenal_cns_high_narrative	1.00
dlbcl_0003_renal_adrenal_cns_high_fhir	1.00
dlbcl_0004_double_hit_fhir	1.00
dlbcl_0004_double_hit_narrative	1.00
dlbcl_0005_limited_stage_gcb_fhir	1.00
dlbcl_0005_limited_stage_gcb_narrative	1.00

v2 task	Reward
dlbcl_0001_bulky_stage_iii_hbv_v2_narrative	1.00

v2 task	Reward
dlbcl_0001_bulky_stage_iii_hbv_v2_fhir	1.00
dlbcl_0002_splenic_stage_iii_cardiac_v2_fhir	1.00
dlbcl_0002_splenic_stage_iii_cardiac_v2_narrative	1.00
dlbcl_0003_journey_v2	5 steps, all 1.00
dlbcl_0003_renal_adrenal_cns_high_v2_fhir	1.00
dlbcl_0003_renal_adrenal_cns_high_v2_narrative	1.00
dlbcl_0004_double_hit_addendum_v2_fhir	1.00
dlbcl_0004_double_hit_addendum_v2_narrative	1.00
dlbcl_0005_limited_stage_flyer_v2_fhir	1.00
dlbcl_0005_limited_stage_flyer_v2_narrative	1.00

Oracle runs prove the pipeline (build → sandbox → answer file → verifier → reward) and that each gold answer scores full marks. They say nothing about model ability.

5. Case-by-case analysis

dlbcl_0001 · DLBCL with nodal disease above and below the diaphragm, splenic lesions and a large retroperitoneal mass: tumor-board plan

Patient	62-year-old man, ECOG 1. Cervical, mediastinal and 11 cm retroperitoneal nodes, splenic lesions, B symptoms.
Teaching point	Spleen is lymphatic tissue, so this is stage III, not IV. Double-expressor by IHC is not double-hit when FISH is negative.
Twist	Hepatitis B serology: HBsAg nonreactive but anti-HBc reactive (resolved/occult infection). HBV DNA not sent; nothing in the chart comments on it.
Key decisions	• Lugano stage III, bulky, B symptoms; IPI 3, CNS-IPI 3 • Pola-R-CHP x6 preferred (R-CHOP x6 acceptable) • Entecavir or tenofovir before the first rituximab dose, continued at least 12 months after (dangerous if missed) • No routine CNS prophylaxis; end-of-treatment PET-CT with Lugano/Deauville
Dangerous items	stage_iii, mn_stage_iv, mn_calls_double_hit, tx_curative_pola_or_rchop, mn_non_curative, hbv_prophylaxis, mn_no_hbv_action
Open questions	• Mandating CNS prophylaxis at CNS-IPI 3 costs 5 points but is not dangerous; the answer still passes at 0.90. • TLS: gold gives allopurinol and hydration; Cairo-Bishop could classify this as high risk, which calls for rasburicase. • HBV monitoring alone, without prophylaxis, is graded dangerous.

dlbcl_0002 · Nodal and splenic DLBCL after prior anthracycline for breast cancer, reduced LVEF: tumor-board plan

Patient	59-year-old woman, ECOG 1. Axillary, mediastinal, mesenteric and para-aortic nodes with splenic nodules; fevers and night sweats.
Teaching point	Splenic involvement with nodes on both sides of the diaphragm is stage III, and the spleen is not an extranodal IPI site (IPI 2).
Twist	Breast cancer in 2016 treated with doxorubicin (240 mg/m² cumulative) and left chest radiotherapy. LVEF has fallen from 62% to 45%, GLS -14%, with no heart-failure symptoms.
Key decisions	• Cancer therapy-related cardiac dysfunction: do not give standard full-dose doxorubicin (dangerous to ignore) • Urgent cardio-oncology review; ACE inhibitor/ARB plus beta-blocker • Curative anthracycline-free regimen: R-CEOP preferred, R-GCVP as alternative; R-CHOP with liposomal doxorubicin or dexrazoxane only with cardio-oncology agreement • Serial LVEF and troponin/NT-proBNP; end-of-treatment PET-CT
Dangerous items	stage_iii, mn_stage_iv, mn_calls_double_hit, tx_cardiac_aware_no_full_dox, mn_standard_dox_ignores_heart, mn_non_curative
Open questions	• Which regimen does the oncologist prefer: R-CEOP, R-GCVP, or cardioprotected R-CHOP? • Should cardiac MRI or an ischemic work-up be required? • Anastrozole: continuing it and holding it are both accepted.

dlbcl_0003 · DLBCL with kidney, adrenal and liver involvement, rising creatinine: tumor-board plan

Patient	67-year-old man, ECOG 2, type 2 diabetes. Retroperitoneal/mesenteric mass with kidney, adrenal and liver lesions.
Teaching point	Stage IV. IPI 5, CNS-IPI 6: kidney/adrenal involvement means high CNS-relapse risk.
Twist	Uric acid rising (9.8 to 10.4), creatinine rising (1.1 to 1.5), LDH more than 5x ULN, taking ibuprofen four times daily, no CNS staging done.
Key decisions	• TLS prophylaxis: hydration, rasburicase after G6PD check, frequent labs, inpatient cycle 1 (dangerous if missed) • Prednisone prephase, then Pola-R-CHP x6 at full dose; G-CSF; stop NSAIDs • MRI brain plus CSF cytology and flow cytometry; HD-MTX optional and deferred until renal recovery • ECOG 2 is caused by the disease, so it is not a reason to reduce doses
Dangerous items	stage_iv, mn_stage_below_iv, mn_calls_double_hit, tx_curative_pola_or_rchop, mn_non_curative, cns_risk_addressed, sup_tls_prophylaxis
Open questions	• R-mini-CHOP for this patient is graded as failing to give full-dose curative therapy (dangerous). Is that too harsh? • Should the steroid prephase be mandatory?

dlbcl_0004 · Large B-cell lymphoma with a late FISH addendum and an outdated R-CHOP plan: tumor-board plan

Patient	54-year-old man, ECOG 1. Right cervical nodes and a 12 cm anterior mediastinal mass narrowing the SVC; above the diaphragm only.
Teaching point	FISH, not morphology or IHC, defines double-hit lymphoma.
Twist	Pathology was signed out as DLBCL NOS and an R-CHOP plan was documented. A later FISH addendum shows MYC and BCL2 rearrangements; the problem list was never updated.
Key decisions	• Revise the diagnosis to HGBL with MYC and BCL2 rearrangements (dangerous if missed) • Cancel the outdated R-CHOP plan; DA-EPOCH-R x6 or a clinical trial • Bulky stage II, IPI 1, but double-hit biology dominates the risk • MRI brain and CSF; CNS prophylaxis considered; mandatory G-CSF; TLS measures; sperm banking
Dangerous items	dx_hgbl_double_hit, mn_dx_dlbcl_nos, mn_stage_iii_iv, tx_intensive_curative, mn_non_curative, mn_cns_dismissed
Open questions	• R-CHOP alone as the preferred regimen costs points but is not dangerous. Is that right for bulky stage II double-hit disease? • An extra must-not (dangerous) was added for dismissing CNS risk because of a low CNS-IPI. Keep it?

dlbcl_0005 · Localized cervical DLBCL in a 45-year-old teacher on a combined oral contraceptive: tumor-board plan

Patient	45-year-old woman, ECOG 0, teacher. Single right cervical nodal region (3.1 cm), no B symptoms.
Teaching point	Limited-stage, low-risk disease, where less treatment is the right answer.
Twist	She has migraine with aura and takes a combined estrogen pill (contraindicated). No pregnancy test is documented. She wants to avoid neck radiotherapy.
Key decisions	• Lugano stage IA, IPI 0, CNS-IPI 0; no marrow biopsy needed after PET • FLYER: R-CHOP x4 plus rituximab x2, no radiotherapy, which also matches her goals • Pregnancy test before treatment; stop the estrogen pill and switch to non-estrogen contraception • Six cycles, Pola-R-CHP or DA-EPOCH-R are overtreatment
Dangerous items	mn_overstaged, mn_calls_double_hit, tx_curative_rchop, mn_observation_or_rt_alone
Open questions	• Six cycles of R-CHOP costs 16 points, enough to fail an otherwise good answer (0.74), but it is not dangerous. Agree? • Missing the pregnancy test and the contraception change together fails the case but is not dangerous. Agree?

6. Longitudinal journey (dlbcl_0003)

Layer	Weight	Situation	Dangerous errors
L1	20%	Diagnosis: the v2 0003 chart	As in the single case (TLS, stage, CNS risk, curative therapy)
L2	15%	End-of-treatment PET Deauville 4; biopsy non-diagnostic; patient declines a repeat biopsy	Calling it a complete response
L3	20%	Progression 3.5 months after Pola-R-CHP; NGS shows EZH2 Y646F	No second-line CAR-T; palliative-only care
L4	30%	Day +6 after axi-cel: confusion, fever, hypotension, ANC 0.2	No steroids for ICANS; no empiric antibiotics
L5	15%	Progression after CAR-T, ECOG 3, declines treatment, financial toxicity	Abandoning the patient

Layers 2-5 are drafts. The L4 differential must include CNS relapse, carried over from the CNS-IPI of 6 at diagnosis.

7. Grader design and known limits

- The regex fallback is a safety net, not the grade of record: it credits keywords anywhere in the answer. Use the judge for any reported number.
- The judge's agreement with an oncologist has not been measured. Target at least 90% item-level agreement on about 30 labelled answers.

- A negation bug (the 'non-' in 'non-bulky' hid a recommendation) was found during the build, fixed, and every case retested.
- Harbor's own multi-step 'mean' averages domain keys with missing values counted as 0. Use scripts/summarize_job.py for weighted journey scores.
- LOINC/SNOMED codes in the v2 builder have not been checked against a terminology server.

8. Next steps

#	Action	Owner	Unblocks
1	Oncologist reviews the 5 case packs and the journey pack (docs/oncologist-review-v2.md)	Oncologist	Any reported v2 score
2	Funded agent key plus a pinned judge model; run configs/harbor-baseline-v1-v2-paid.json (3 attempts each)	Engineering	First trustworthy baseline; v1 vs v2 and narrative vs FHIR
3	Judge calibration against about 30 oncologist-labelled answers	Engineering + oncologist	Trust in judge scores
4	Oncologist completes journey layers 2-5 on the longitudinal form	Oncologist	Journey scores
5	Terminology validation of FHIR codes	Engineering	External delivery
6	Scale to about 20 patients using the cases_src template; second tumour type	Engineering + oncologist	Pilot package

9. Where things are

Path	Contents
reports/cases/	One confidential case pack PDF per v2 case, plus the journey
cases_src/	Single source per patient: case.toml (visible), gold.json, rubric_v2.toml, selftest/ (hidden)
graders/oncoworld_grader/	Shared v2 grader, test.sh, self-test runner
scripts/build_cases_v2.py, build_journey_v2.py	Generate the Harbor tasks; enforce the leak, consistency and gold checks
scripts/summarize_job.py	Per-case, per-version and journey-weighted summaries of a Harbor job
docs/phase2-clinical-realism.md	Engineering documentation and commands
docs/oncologist-review-v2.md	Sign-off questionnaire

Synthetic cases for AI evaluation only. Not clinical decision support. No real patient data.